



Beyond technique: the art of writing impactful econometric papers for information technology for development

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EDITORIAL



Beyond technique: the art of writing impactful econometric papers for information technology for development

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ABSTRACT

This editorial provides methodological guidance for researchers publishing econometric studies in *Information Technology for Development (ITD)*. Successful econometric research requires more than sophisticated techniques, it demands theoretical grounding and careful research design. The paper follows an hourglass structure, addressing research positioning, theoretical framing, and sample selection. Econometrically, I demonstrate how minor changes in model specification can dramatically change a paper's results, particularly with Generalized Method of Moments (GMM) estimators in dynamic panel data. I highlight some recent examples of strong econometric papers in *ITD*. Researchers in information systems and development economics will find practical strategies for strengthening their econometric arguments and producing more impactful research.

1. Introduction

In my role on the editorial board of *Information Technology for Development*, I've had the privilege of guiding and evaluating many research papers. Certain issues arise frequently, and I'd like to address some common areas where submissions often need improvement, particularly regarding data, estimation, and econometrics.

Academic papers should follow an hourglass structure. Begin with compelling big-picture questions, narrow the focus through the literature review, and further concentrate on your specific dataset and estimation results. Then gradually broaden your perspective, connecting your findings to larger questions and discussing potential policy implications of your work.

In what follows, I'll look at each section of a research paper and comment its specific relevance to the econometrics section. I'll specifically address some complications of the popular Generalized Method of Moments (GMM) framework for dynamic panel data (notice we're getting quite specific here). We'll examine at each part of a paper in sequence. The structure below should look familiar!

2. Research question

Good research begins with an interesting and important question. As an editor, I must read each paper that comes across my desk. Professional ethics requires me to read every word, weighing whether a paper has the potential to make a significant impact in the field, and whether it can be improved to the point of publication. Your readers—your real readers, not the referees—are

under no such obligations. You must convince them immediately that your paper is worth reading and that you have something interesting to say. Is your research question timely and interesting? It must be compelling enough that your readers—fellow academics with teaching responsibilities and research agendas—invest their valuable time in reading your paper rather than someone else's.

Research questions need proper framing. Demonstrate that your question has broad relevance, addressing larger-scale issues that might interest researchers beyond *ITD*. The field of Economic Development has several long-running debates your research could engage with: Does your work provide evidence for endogenous versus exogenous growth theory? Does it help determine whether factors such as capital, can be fruitfully imported into low and middle income countries, or whether capital is institutionally specific? Is information technology a mechanism for growth that generates spillovers, increasing incomes broadly rather than just for direct users? What characteristics make certain technologies potential general-purpose tools that drive economic growth?

Fundamental questions about endogeneity versus exogeneity, exportable growth, and the relative importance of physical, legal, or social infrastructure have persisted for decades. Don't be afraid to connect your research to these larger historical questions; doing so makes your work relevant to scholars outside the Information Technology for Development discipline and strengthens its impact.

Using a novel technique is not a research question. Using a new dataset is not a research question. Perhaps the topic has been under-studied. Perhaps it has been studied with an insufficient methodology. Establish this in broad strokes, and only in broad strokes, in the introduction.

3. Literature review

The literature review allows authors to distill extensive research for their readers' benefit. Your goal isn't to impress with citation counts, but to help readers understand the main issues and approaches relevant to your research question. What *patterns* have emerged from previous research? For example, you might identify that one group of researchers believes ICT breaks the link between output and pollution, while another group argues it doesn't. Do different researchers reach different conclusions? Do these differences stem from varying methodologies or sample selections? Thematic grouping is essential—a mere list of papers and their findings makes for neither useful nor compelling reading and suggests the author cannot distinguish between significant patterns and individual studies.

Which papers should you cite? Aim for relevance rather than exhaustiveness. The purpose of a literature review is to bring readers up to speed on the general progression of the research stream. Group papers thematically to help readers comprehend the broader research landscape rather than individual studies. Most literature reviews would benefit from clearer structure. Not all papers carry equal weight—help your readers navigate by focusing on significant contributions and important conclusions drawn from groups of papers. If you find yourself citing numerous poorly-cited papers on your topic, consider whether this research stream has run dry. Without establishing a truly novel approach to well-trodden territory, your paper will likely have minimal impact and face publication challenges. Similarly, if your bibliography exceeds 100 citations, either many papers can be eliminated as tangential to your research question, or your question lacks novelty and will struggle to make an impact. An extensive review covering over 100 papers may indicate little remains unsaid about your topic. Instead, develop a thematic analysis of major findings in the literature.

Finally, your literature review serves as a bridge to the narrower portions of the paper, namely the data and econometric sections. Although they serve distinct purposes, the literature review and the empirical sections are not independent of each other. Your literature review should indicate which variables are important and how they are related. The literature review can even discuss the econometric models that have been used by other researchers, especially if you believe those models are incomplete or misleading.

4. Theory

Well-written literature reviews naturally explore evidence for or against different theories. As Nobel Prize-winning economist Tjalling Koopmans (1947) once said, “estimation without theory is data mining.” In this section, you must clearly articulate the particular theory you’re testing or the theoretical constructs you’ll be using.

Many macro-econometric papers submitted to *ITD* follow a common pattern. Typically, the dependent variable is income, health, education, or another UN development target. The usual independent variables are mobile access, broadband access, or internet access (often all three), along with some control variables. This structure has unfortunately become standard. Recent papers often attempt to establish originality by including an interesting “third variable”—perhaps education, institutions, or rule of law—frequently interacted with the IT variables. The resulting paper is then titled “The IT/Education/Income Nexus.” That word “nexus” often signals limited theoretical development. If your model includes interaction terms, you must provide a solid theory why the marginal impact of *X* would depend on *Z*. It is insufficient to merely hypothesize that such a relationship might exist. Ideally, your work should build on well-established theories developed outside your current paper.

A path diagram can effectively summarize theoretical relationships. However, simply recasting your regression equation as a path diagram doesn’t make it a theory. The theoretical framework must precede your empirical strategy.

Sometimes it is evident that the theory section has been added retroactively to rationalize empirical findings. The scientific process begins with an interesting research question or hypothesis, followed by data collection to test that theory. The theory comes first. Post-hoc theorizing inverts this process (Kerr, 1998; Leung, 2011). It is too easy to rationalize random results after the fact. Such post-hoc theorizing is unscientific, akin to astrology, where theories are crafted to fit the data before you.

To quote Lord Kelvin, “Theory without measurement is mere speculation.” And so we now turn to the data and econometrics sections.

5. Data

Macro research designs typically involve the choice of using single-country time-series, regional panel data, or global panel data. Each approach has implications for external validity—a crucial consideration when selecting your data.

Papers titled “ICT and *X*: the case of Genovia,” or something like that, raise an important question: Why is this specific case study worthy of readers’ attention? Authors must explain why their chosen country is *particularly* rather than *coincidentally* relevant. To establish external validity, clarify how lessons from studying Genovia can be generalized beyond its borders.

Case studies become valuable when they either (1) offer unique insights unavailable elsewhere, or (2) examine sufficiently generic contexts where results can generalize. Perhaps Genovia is the only country to collect a particular type of data or to have tried a specific policy experiment. Alternatively, Genovia might represent an “average” country, making it a meaningful proxy for similar nations. Without establishing this relevance, scholars and policy makers from other countries will find little reason to apply your findings to their situations, and your paper risks being dismissed as too narrowly focused.

Collecting data across multiple countries helps address some concerns about generalizability. Panel data allows researchers to report average correlations across various nations, strengthening external validity. However, this approach comes with trade-offs. By aggregating across countries, you may overlook important nuances and context-specific factors that could contain valuable information. Additionally, most panel data techniques involve numerous hidden assumptions that your

data may or may not satisfy. The popular Arellano-Bond class of Generalized Method of Moments (GMM) estimators exemplifies these challenges—a topic we’ll explore shortly.

6. Regression and estimation

Regressions serve to test or illustrate theories—they are never atheoretical, and data rarely “speak for themselves.” If your theory is well specified, the functional form of your regression equations and estimation methodologies should follow naturally from your theoretical framework.

Many researchers fall into the “spaghetti model” trap—throwing variables against the kitchen cabinet to see what sticks. Just as some test pasta’s doneness by seeing if it adheres to a cabinet (a practice my Italian colleagues would find horrifying), researchers sometimes gather various variables to see which regression produces desired results. This constitutes a form of *p*-hacking. Post-hoc theorizing is often identifiable by poorly articulated explanations of results. It’s much harder to *p*-hack when strong theoretical reasons connect your variables causally.

Your estimation methodology must be econometrically sound. For single-country macroeconomic case studies, follow a systematic workflow: First, test for stationarity and proceed accordingly: use ordinary least squares for stationary variables; autoregressive distributed lag or Toda and Yamamoto (1995) models when variables have mixed stationarity properties; and test for cointegration when dealing with first-difference stationary variables. For panel data, be mindful of Cross-Sectional Dependence (CSD), which occurs when error terms across different units (e.g., between countries) are correlated due to common shocks or economic interdependence. Several diagnostic tests can detect and accommodate CSD in panel data models. Chapter 13 in Leventis (2023) textbook, *Time Series Econometrics, 2nd ed*, provides an introduction to these methodological considerations.

Your choice of estimator should connect closely to your research question and available data. It sometimes appears that papers proliferate whenever Stata updates its menu of estimators. Before deploying sophisticated methods, carefully establish why they are necessary. Justify why your particular estimator best investigates your research question. Papers often estimate new models—threshold models, quantile regressions, or include interaction effects—without adequate justification. Articulate a theoretical connection between your variables and methods to avoid accusations of data mining or using exotic estimators merely for novelty.

A research paper is as much an exercise in rhetoric and argumentation—trying to convince someone of the correctness of the methods and conclusion—as it is an exercise in science (McCloskey, 1983). As Deirdre McCloskey put it, “rhetoric is disciplined conversation” (p. 483). A sound econometric methodology provides this discipline. Authors are encouraged to establish why the methods are necessary and appropriate.

Further, much like a picture is more powerful than a hundred words, a simple model is often more convincing than a more complicated one. As Albert Einstein was reported to have said, “Make everything as simple as possible, but no simpler.” When in doubt, use simpler more easily interpretable models. Ultimately, your econometric methods should only be as complex as required. Complicated methods frequently yield results that are sensitive to hidden and subtle modeling assumptions. These assumptions often remain unstated and unverified.

Establish that your results are robust. Small changes in model specification—such as including or excluding particular control variables—can dramatically alter estimated coefficients. Traditionally, authors present slight variants of their models in the same table to demonstrate result stability across specifications.

Finally, coefficients should make sense. When coefficients on related variables like mobile and broadband access have opposite signs, question whether there’s a strong theoretical explanation. Isn’t broadband access simply a means to internet access? A sign change represents an important qualitative difference, not merely a quantitative one—but is this a genuine result or a Type-I error? Sign changes often indicate multicollinearity or model misspecification. Authors need to articulate a compelling theoretical reason for such sign differences; otherwise, all results become

suspect. Without a convincing causal story explaining these contradictory patterns, correlations remain mere numbers, and readers will rightfully question the validity of your entire analysis.

6.1. GMM estimators

If you're estimating a dynamic panel data model, don't uncritically opt for one of the Arellano-Bond family of Generalized Method of Moments (GMM) estimators. These estimators are quite popular but they rely on assumptions and are only optimal when those assumptions are met. For example, the GMM estimators assume no cross-sectional dependence (CSD). If you suspect CSD, consider using the dynamic panel-data estimator of Chudik and Pesaran (2015).

For GMM estimators to work well they require that the number of cross-sectional units (N) is much larger than the number of time periods (T).

Most dynamic panel data models (including GMM) involve estimating in first-differences. This, however, introduces endogeneity which in turn creates makes standard estimators biased (Nickell, 1981; Levendis, 2023). GMM addresses the endogeneity bias by using instrumental variables. Good instruments are hard to find. One of the conveniences of the Arellano-Bond framework, though, is that the instruments are generated from the dataset itself.

While instrumental variables methods correct for endogeneity, they are notoriously unstable. Using slightly different instruments can result in dramatically different estimates.

Weak instruments do not remove the endogeneity bias. And having too many instruments can re-introduce the endogeneity bias. As T increases, the number of instruments grows quadratically. This implies that the instruments can overfit the endogenous variable and fail to filter out the correlated error term. In other words, having too many instruments can be as bad as having no instruments at all. Roodman's (2007) paper, "A note on the theme of too many instruments" offers a deep exploration on GMM's instrumentation problems and possible solutions. A common rule of thumb is to keep the number of instruments smaller than N . Practitioners often "collapse" the instrument matrix or limit the lag depth to mitigate these problems when T is large.

There are two "flavors" of GMM estimators, difference-GMM introduced by Arellano and Bond (1991), and system-GMM by Arellano and Bover (1995) and Blundell and Bond (1998). The methods differ in their estimating equations and instrument sets.

The difference GMM approach uses lagged levels as instruments for differenced equations. Lagged levels are often weak instruments for first differences, especially when the series is highly autocorrelated. This can lead to finite sample bias and poor precision. It works poorly in small samples, especially with persistent data, and finite sample bias is substantial when N is small.

The system-GMM method was developed specifically to address difference-GMM's weak instrument issue by using both lagged levels as instruments for differenced equations and lagged differences as instruments for level equations. System-GMM generally has better finite sample properties than difference-GMM when series are persistent. However, it requires additional assumptions about initial conditions and can be more sensitive to violations of assumptions in small samples. The centrality of this is revealed by the title of Blundell and Bond's (1998) paper: "Initial conditions and moment restrictions in dynamic panel data models."

Regardless of which version of GMM you use, essential diagnostic tests must be reported, including the Sargan and Hansen tests for overidentifying restrictions and the Arellano-Bond test for autocorrelation in the differenced errors. It should be noted, however, that weak instruments can also reduce the statistical power of the Hansen and Sargan tests, potentially leading to falsely accepting those same weak instruments.

In summary, econometric results can be highly sensitive to the choice of instruments, lag structure, and transformation method (differences vs. system approach). Referees and readers should be able to exactly replicate your results. Especially if using GMM, you need to list exactly which variables are being instrumented, how they are being instrumented, whether the instrument list is being collapsed, and any other options that you've specified.

6.2. GMM results

In this section I'll illustrate GMM's sensitivity to method, control variables, and endogeneity/exogeneity assumptions. We will see how dramatically a paper's conclusions can depend on seemingly trivial assumptions and modeling choices.

We'll investigate how IT and education affect average income in a panel of Sub-Saharan countries for the thirty years spanning 1995–2024. All of our data are standard: *GDP* is real GDP per capita, measured in 2010 US dollars. *Bandwidth* is measured by International Internet bandwidth (bits per person). *FDI* is net inflow of FDI measured as a percentage of GDP. *Illiteracy* is the number of adults in the population who are illiterate (measured in millions). All data are from the World Bank and International Communication Union via the *wbopendata* package in Stata.

We begin by estimating the dynamic panel data model described by Equation (1). The equation contains GDP as the lagged dependent variable. We include an IT variable (*bandwidth*) and a control variable (*illiteracy*). Fixed effects are included via the index-specific error variable (u_i). Each observation is hit by an error term which, by assumption, has no cross-sectional dependence.

$$GDP_{i,t} = \beta_0 + \beta_1 GDP_{i,t-1} + \beta_2 Illiteracy_{i,t} + \beta_3 Bandwidth_{i,t} + u_i + \varepsilon_{i,t} \quad (1)$$

Table 1 reports four different estimates of Equation (1). In column (1) I use the difference GMM estimator of Arellano and Bond. All variables are statistically significant at the 0.01 level.

The Hansen p -values are high (above 0.05) indicating that the instruments are exogenous. (The instances where the Hansen p -values are equal to one indicates too much of a good thing: too many instruments.) Some Sargan p -values are zero indicating that the instruments are endogenous.) Thus, our Hansen and Sargan p -values indicate different conclusions an endogeneity/exogeneity.

The coefficients in column (1) reveal that real per capita GDP in sub-Saharan Africa does not have a high degree of persistence. A one unit increase in the illiteracy variable (indicating one million more illiterate adults in the country) corresponds with an increase in real yearly incomes by \$5.64 US dollars. Most importantly for a journal like *ITD*, we see that the IT variable, bandwidth, is also statistically significant at the 0.01 level. A 1Kb increase in bandwidth is associated with an increase in annual incomes of \$69 per person. But should we care? Do you trust this result? The coefficient on illiteracy was positive. By this logic, making an entire country illiterate would be the key to economic development! No. This indicates that the model is probably misspecified. Pay attention to those coefficients!

As mentioned earlier, GMM estimators can be very sensitive to various subtle modeling choices. In columns (2)–(4) we estimate Equation (1) using the system GMM estimators. We are estimating the

Table 1. Dependent variable: real per capita GDP.

	(1) Diff-GMM	(2) Sys-GMM	(3) Sys-GMM	(4) Sys-GMM
Lag.GDP	0.186*** (0.000)	0.932*** (0.000)	0.930*** (0.000)	0.931*** (0.000)
Illiteracy	5.642*** (0.000)	0.059 (0.988)	-0.340 (0.910)	0.217 (0.959)
Bandwidth	0.069*** (0.000)	0.116** (0.035)	0.117** (0.034)	0.118** (0.031)
_cons	1217.985*** (0.000)	95.677 (0.109)	100.658* (0.058)	95.978* (0.100)
<i>N</i>	86	86	86	86
AR(1) p -value	0.316	0.381	0.384	0.381
AR(2) p -value				
Hansen p -value	0.772	1.000	1.000	1.000
Sargan p -value	0.037	0.000	0.000	0.000

Notes: Columns (1) and (2): Illiteracy as exogenous; GDP and bandwidth as endogenous. Column (3): Illiteracy and bandwidth as exogenous; GDP as endogenous. Column (4): GDP, illiteracy and bandwidth as endogenous.

p -values are in parentheses; * $p < .10$, ** $p < .05$, *** $p < .01$.

exact same equation, with a slightly different version of same estimator. But we get very different results. To be clear, the three sys-GMM estimators are fairly consistent with each other, but their coefficients are very different from the diff-GMM estimates' coefficients. First, all three sys-GMM estimators find extremely high persistence for real per capita GDP. In fact, they are almost equal to one, nearly making them unit-roots! This is in direct contrast with column (1). The marginal effect of a 1Kb increase bandwidth is approximately \$117, nearly double the estimate from column (1). Wow! The sys-GMM estimators find that, after controlling for the effect of IT via bandwidth, illiteracy is insignificant. IT is more important than literacy! No, these models are probably also misspecified.

Also, notice that models (2)-(4) are all sys-GMM estimates of Equation (1). So why are they different? The three models make different assumptions about which variables are treated as endogenous and exogenous. Ultimately in these models, the choice of endogenous vs exogenous seems not to have mattered too much. But it often does, and many papers skim over their explanations for why they treat some variables as endogenous and others as exogenous.

Table 2 also compares a diff-GMM estimator and three sys-GMM estimators, but with Foreign Direct Investment (FDI) added as a control variable. Thus, the four columns are estimates of Equation (2):

$$GDP_{i,t} = \beta_0 + \beta_1 GDP_{i,t-1} + \beta_2 Illiteracy_{i,t} + \beta_3 Bandwidth_{i,t} + \beta_4 FDI_{i,t} + u_i + \varepsilon_{i,t} \quad (2)$$

Column (1) of Table 2 has all variables as statistically significant at the 0.01 level. Illiteracy is a strong determinant of real per capita income. Naturally. But notice that its coefficient is positive and very, very large. According to the estimates in column (1), every million additional illiterate adults in the economy increases average incomes by \$191. Wow; ignorance truly is bliss. Bandwidth increases average incomes by \$52 per Kb. FDI is significant and negative, implying that foreign investment is far from beneficial. Instead, this is evidence that colonialism is harmful, extracting resources and lowering incomes. This is post-hoc theorizing.

Columns (2)-(4) of Table 2 display the sys-GMM estimates. Column (1) had an unreasonably large positive value for illiteracy. The coefficients on illiteracy from columns (2)-(4) are much smaller, hovering around 1. These are still positive though, implying that more illiteracy is good for the economy. No, they are more evidence of misspecification. Interestingly, the effect of FDI went from negative and significant in column (1) to positive and insignificant in columns (2)-(4). But we estimated exactly the same equation? Why the difference?

Table 2. Dependent variable: real per capita GDP (with FDI as control).

	(1) Diff-GMM	(2) Sys-GMM	(3) Sys-GMM	(4) Sys-GMM
Lag.GDP	0.467*** (0.000)	1.019*** (0.000)	1.019*** (0.000)	1.019*** (0.000)
Illiteracy	190.616*** (0.000)	0.931*** (0.003)	1.096*** (0.000)	0.931*** (0.003)
FDI	-48.524*** (0.000)	0.731 (0.318)	0.419 (0.282)	0.731 (0.318)
Bandwidth	0.052*** (0.000)	-0.006 (0.401)	-0.005 (0.412)	-0.006 (0.401)
_cons	0.000 (.)	-4.193 (0.468)	-3.919 (0.443)	-4.193 (0.468)
N	73	73	73	73
AR(1) p-value	1.000	0.403	0.413	0.403
AR(2) p-value	.	.	.	.
Hansen p-value	1.000	1.000	1.000	1.000
Sargan p-value	0.035	1.000	1.000	1.000

Notes: Columns (1) and (2): Illiteracy and FDI as exogenous; GDP and bandwidth as endogenous. Column (3): Illiteracy and bandwidth as exogenous; GDP and FDI as endogenous. Column (4): FDI as exogenous; GDP, illiteracy and bandwidth as endogenous. p-values are in parentheses; *p < .10, **p < .05, ***p < .01.

The columns in [Table 2](#) differ, not in the equation they are estimating, but in the hidden assumptions that allow for the estimation. Columns (1) and (2) assume that bandwidth is endogenous with income, and that illiteracy and FDI are exogenous. This needs to be justified. I think a case can be made that bandwidth is endogenous: increased bandwidth can increase incomes, and higher incomes can buy more bandwidth. Column (3) assumes that FDI is endogenous, and illiteracy and bandwidth are exogenous. Column (4) assumes that illiteracy and bandwidth are endogenous, and FDI is exogenous. For technical reasons, it turns out that columns (2) and (4) are identical models even though their list of endogenous and exogenous variables are different.

Finally, notice how adding a single control variable significantly alters the conclusions. For example, comparing the diff-GMM estimators in the first columns of [Tables 1](#) and [2](#), we see that adding FDI changed the estimate of Illiteracy's impact from 5.64 to 190.62, a 33-fold increase. It also leads to a complete reversal in the Sargan p -statistics for the sys-GMM estimates from 0.000 to 1.000. Small changes in modeling assumptions can have a huge impact on a model's diagnostics and conclusions.

7. Recommendations and examples

Constructive criticism is most valuable when paired with positive examples. Several recent publications in *Information Technology for Development* demonstrate the various qualities I've advocated throughout this editorial.

The paper by Emilie Bonhoure ([2025](#)) is an excellent example of how to position econometric research within a broader scientific debate and integrate theory throughout a paper. While Bonhoure investigates a seemingly narrow topic—the correlation between indigenous knowledge and digital financial inclusion (DFI)—its broader contribution is exploring the relative importance of formal and informal institutions. The paper extends Douglass North's Nobel-Prize winning work in institutional economics, examining how formal structures (like laws) and informal ones (like customs) can lower transactions costs. Bonhoure's econometric models logically include formal and informal variables. Bonhoure ([2025](#)) contrasts two groups of countries: Europe, with strong formal institutions, and Sub-Saharan Africa (SSA), with weaker formal institutions but robust informal ones. Bonhoure suggests that M-Pesa's (the mobile payments platform) development in Kenya is no coincidence. Formal institutions increase DFI in Western Europe, where formal institutions are strong. More intriguingly, formal rules negatively correlate with DFI in SSA, where informal institutions dominate. Bonhoure demonstrates that formal and informal institutions act as substitutes and, more controversially, suggests a potential asymmetry where informal rules may actually be superior. These insights speak to economists, political scientists, and sociologists, exemplifying how a well-positioned paper can contribute to wider academic conversations.

There are many ways to organize a literature review thematically. The literature review in Zhao et al. ([2025](#)), for example, is organized into pairs: (a) digitalization and SME performance, (b) dynamic capabilities and SMEs. This clean division helps readers quickly grasp the landscape of existing research.

Mulungu et al. ([2025](#)) conduct a systematic review of the literature on ICT and agricultural-extension programs. This field presents a particular challenge—studies vary widely in the programs they examine, countries they focus on, crops they consider, and extension program specifics. It becomes essential to be able to sort the wheat for the chaff, to find the common kernels of insight between the various studies. Mulungu et al. categorize and then summarize the conclusions of 49 recent papers. They find that ICT increases awareness of good agricultural techniques, changes agricultural practices, increases yields, and increases outcomes. Their review goes beyond mere summary to offers concrete policy suggestions: the most effective ICT agricultural-extension programs are carefully tailored to user needs, have well-trained users or include training programs, are compatible with existing services, and feature interactive platforms that enable discussion rather than one-way communication like SMS advisories.

Roberts and Oosterom (2024) demonstrate how to position research for broader relevance by starting with a widely recognized trend: the global rise of authoritarianism and decline of democracy. This framing immediately appeals to scholars across political science, communications, IT, and law. They then narrow their focus, exploring the literature on authoritarianism and the digital world. Their thematic organization is historical, pointing out that the field was once concerned with the digital practices of authoritarian regimes across the world, but more recently has studied the authoritarian digital practices by otherwise non-authoritarian countries. The authors methodically explore various definitions, showing how different studies align with each conceptual approach. Readers are left with an understandable road map by which to navigate the broad and evolving literature.

Your sample and method fit the research question. Alotaibi et al. (2025) show this well in their study of whether mobile government services (M-gov't) increase women's autonomy and freedom. Their approach—interviewing 30 Saudi women—is appropriate for exploratory research. Their selection of Saudi Arabia as a case study is particularly well-justified. The country has historically been an outlier regarding women's rights, but recent exogenous changes in guardianship laws allow women to access some government functions remotely. The exogeneity of this change allows the authors to make a causal claim: M-gov't increased the freedoms and capabilities of Saudi women. For example, one participant grew her business by adding delivery services once she could drive. Saudi Arabia's unique position makes the resulting changes more visible and measurable than they might be in countries with already-established women's freedoms. Furthermore, given Saudi Arabia's considerable influence among Middle East and North Africa countries, this study provides a valuable test case to predict how similar changes might affect women across the region. This thoughtful selection of research context demonstrates how a carefully chosen sample can enhance both the internal validity of findings and their potential for broader generalization.

Li et al.'s (2025) work on the digital divide and subjective feelings of deprivation offer an example of good econometric writing. They organize their literature review along three levels: differences in access, usage, and utility. Their theory section establishes that information is crucial for good decision making, and information is often exchanged informally via social networks. Their hypotheses are explicitly stated, tie cleanly to their theoretical lens, and are succinctly summarized by a path diagram. In their regression tables, control variables are thoughtfully grouped into three categories: (a) individual demographic statistics, (b) household variables, and (c) regional characteristics. They conduct various statistical robustness tests and establish the consistency of their results across different variations of their base model. Readers are thereby more confident in accepting their conclusions: usage and utility gaps increase feelings of deprivation, undermine social networks and reduce social trust.

Each of these papers shows how econometric technique is not a substitute for clear thinking, proper positioning, and good writing.

8. Conclusions

Information Technology for Development receives hundreds of submissions each year. Many promising papers have to be rejected. But there are some things authors can do before submitting to increase the chances that the paper will be published. Establish an interesting, important, or novel question. Relate this to an established theory. Explain, in general terms, what the scientific community has already discovered about the question. Don't get bogged down in individual papers but focus on the overall landscape. Explain why you chose a particular country or subset of countries for your analysis. Use an appropriate dataset for the research question, and an appropriate econometric model for the theory and data. Finally, summarize and offer some suggestions. (Just like I did just now!)

Notes on contributor

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